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Q1. Program based on Classes and Objects

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Student {
    public int rollNo;
    public String name;
    public double marks;
    public void displayDetails() {
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
        Student s1 = new Student();
        int roll = in.nextInt();
        in.nextLine();
        String name = in.nextLine();
        double marks =in.nextDouble();
        s1.rollNo = roll;
        s1.name = name;
        s1.marks = marks;
        s1.displayDetails();

    }

}
```

INPUT

```
101
Dharanish
99.5
```

OUTPUT

```
Roll No: 101
Name: Dharanish
Marks: 99.5
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student {
    private int rollNo;
    private String name;
    private double marks;
    void setRollNo(int rollNo)    {
        this.rollNo = rollNo;
    }
    void setName(String name) {
        this.name = name;
    }
    void setMarks(double marks) {
        this.marks = marks;
    }
    void displayDetails() {
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        int roll = sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        s.setRollNo(roll);
        s.setName(name);
        s.setMarks(marks);
        s.displayDetails();
        sc.close();
    }


}
```

INPUT

```
101
dharanish
99.8
```

OUTPUT

```
Roll No: 101
Name: dharanish
Marks: 99.8
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Student {
    int rollNo;
    String name;
    double marks;
    Student() {
        rollNo =101;
        name ="Rahul";
        marks=85.5;
    }
    Student(int rollNo, String name, double marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }
    void displayDetails() {
        System.out.println("Roll No: " +rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }


    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student s1 = new Student();
        s1.displayDetails();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        Student s2 = new Student(rollNo, name,marks);
        s2.displayDetails();
        sc.close();
    }
}
```

INPUT

```
101
rahul
85.5
```

OUTPUT

```
RollNo: 101
Name: Rahul
Marks: 85.5
RollNo: 101
Name: rahul
Marks: 85.5
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Employee {
    int empid;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int empid, String empName, double basicSalary) {
        this.empid = empid;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int empid = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();
        Salary s =new Salary();
        s.getEmployeeDetails(empid, empName, basicSalary);
        s.calculateSalary();
        s.displayDetails();
        sc.close();
    }
}

class Salary extends Employee{
    double hra, da, netSalary;
    void calculateSalary() {
        hra = basicSalary*0.20;
        da = basicSalary*0.10;
        netSalary = basicSalary+hra+da;
    }
    void displayDetails() {
        System.out.println("Employee ID: "+ empid);
        System.out.println("Employee Name: "+ empName);
        System.out.println("Basic Salary: "+ basicSalary);
        System.out.println("HRA: "+ hra);
        System.out.println("DA: "+ da);
        System.out.println("Net Salary: "+ netSalary);
    }
}
```

INPUT

```
101
Rahul
25000
```

OUTPUT

```
Employee ID: 101
Employee Name: Rahul
Basic Salary:25000.0
HRA: 5000.0
DA:2500.0
Net Salary:32500.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee {
    int empId;
    String empName;
    void getEmployeeDetails(int empId,String empName) {
        this.empId = empId;
        this.empName = empName;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();
        GrossSalary obj = new GrossSalary();
        obj.getEmployeeDetails(empId, empName);
        obj.getBasicSalary(basicSalary);
        obj.calculateSalary();
        obj.displayDetails();
        sc.close();
    }
}
class Salary extends Employee {
    double basicSalary;
    void getBasicSalary(double basicSalary)
    {
        this.basicSalary = basicSalary;
    }
}
class GrossSalary extends Salary {
    double hra, da, grossSalary;
    void calculateSalary() {
        hra = basicSalary*0.20;
        da = basicSalary*0.10;
        grossSalary = basicSalary + hra + da;
    }
    void displayDetails() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Gross Salary: " + grossSalary);
    }
}
```

INPUT

```
102
Dharanish
50000
```

OUTPUT